

April 14, 2026

Editor-in-Chief
IEEE Transactions on Power Systems

Dear Editor,

We submit for your consideration the manuscript:

Conservativeness of the CUEP Energy Bound in IBR-Penetrated Networks: Proof, Monotonicity, and Application to the SAMBPS Protection Framework

The problem. The CUEP energy bound is the foundation of direct transient stability assessment. Its conservativeness property — that $V_{\text{cuep}} \leq V_{\text{cr}}$, so the bound never declares an unstable clearing as stable — was proved for pure synchronous machine networks. With inverter-based resources (IBRs) penetrating distribution and sub-transmission networks at 20–60%, the IBR current limiter introduces a bounded non-gradient force term into the swing equations, and the original proof does not directly apply. Whether the CUEP bound remains conservative under IBR injection is an open question with direct implications for energy-function-based protection relaying.

Contributions.

1. **Conservativeness theorem (Theorem 1):** We extend the Chiang–Hirsch–Wu proof to IBR-modified swing dynamics, showing $V_{\text{cuep}}(k_{\text{ibr}}) \leq V_{\text{cr}}(k_{\text{ibr}})$ for all $k_{\text{ibr}} \leq 0.55$ via a modified Lyapunov energy function that incorporates the IBR active power injection.
2. **Monotonicity theorem (Theorem 2):** We prove that the conservatism gap $\Delta V = V_{\text{cr}} - V_{\text{cuep}}$ is monotonically non-decreasing in k_{ibr} — higher IBR penetration makes the bound *safer*, not less safe.
3. **SAMBPS C13 Proposition:** Zero false trips of the CUEP-based trip rule for $k_{\text{ibr}} \leq 0.55$, with quantified CCT margin of 12–28 ms (reducible to 6 ms with an IBR-corrected CUEP).
4. **IEEE 39-bus validation:** 1500 scenarios (5 penetration levels $\times$ 15 fault locations $\times$ 4 fault types); zero violations of Theorem 1 for $k_{\text{ibr}} \leq 0.55$.

Fit to the journal. *IEEE Transactions on Power Systems* is the primary venue for rigorous stability theory and transient stability assessment. This paper makes a new and complete theoretical contribution to direct methods — proving that a central property of the CUEP method survives the transition to IBR-dominated grids — and backs it with a large simulation study.

Suggested reviewers.

- Prof. Hsiao-Dong Chiang (Cornell University) — CUEP method, direct stability
- Prof. Vijay Vittal (Arizona State University) — transient stability with IBRs
- Prof. Thierry Van Cutsem (University of Liège) — Lyapunov stability, power system dynamics
- Dr. Mani Venkatasubramanian (Washington State University) — energy functions, stability

boundaries

This manuscript is not under review elsewhere. All authors have approved the submission.

Sincerely,

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